

Lane Centering Under Camera Failures: Classical Control vs. Reinforcement Learning for ADAS

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Abstract—Lane centering systems in modern vehicles rely primarily on forward-facing cameras for lane detection, rendering them vulnerable to sensor failures from occlusion, adverse weather, and hardware degradation. This paper presents a controlled simulation framework comparing a classical Kalman filter paired with a nested PID controller against a Soft Actor-Critic agent with an LSTM actor (SAC-LSTM) for autonomous lateral lane centering under camera sensor failures. Both controllers are evaluated on a shared MATLAB/Simulink plant model of a Cadillac LYRIQ (2600 kg, 3.094 m wheelbase, 15 m/s) across five road scenarios and six Markov chain camera failure states (F0–F5). The classical baseline achieves sub-2 cm RMS lateral error under nominal conditions with graceful degradation under stochastic and burst dropout, and exhibits a well-defined failure boundary under sustained 3s dropout on curved geometry. SAC-LSTM training across thirteen experimental runs characterizes a fundamental tracking-smoothness tradeoff in the reward structure, where agents achieving strong tracking metrics do so via high-frequency steering oscillation rather than smooth corrective actuation. The shared-plant evaluation framework, Markov chain failure model, and validated classical baseline constitute the primary contributions of this preliminary work, establishing the foundation for a rigorous head-to-head comparison as SAC-LSTM training progresses.

Index Terms—lane centering, autonomous driving, reinforcement learning, Kalman filter, PID control, sensor failure, ADAS, SAC, LSTM, camera dropout

I. INTRODUCTION

Lane centering systems have become standard equipment across vehicle segments, transitioning from luxury features to widespread production deployment. Classified as Level 1 or Level 2 automation under SAE J3016 taxonomy [1], these systems actively maintain vehicle position within lane boundaries through continuous steering correction. As the automotive industry progresses toward higher automation levels, the robustness of these foundational lateral control systems becomes increasingly critical for both occupant safety and public acceptance of autonomous technologies. Production implementations rely predominantly on forward-facing cameras

for lane marking detection, paired with classical architectures combining Kalman filtering for state estimation with nested PID controllers for steering actuation [2], [3]. This camera-centric approach offers the spatial resolution necessary for lane detection and remains cost-effective for mass-market deployment [4].

This dependence on camera perception introduces a fundamental vulnerability. Camera failures span hardware-level faults through environmental degradation [5]–[7]—including brightness saturation, occlusion, condensation, and adverse weather—each producing distinct degradation signatures in lane detection output that propagate directly to lateral tracking error with no graceful fallback in systems lacking redundant perception modalities.

Two paradigms have emerged to address sensor robustness in lane-keeping systems. Classical methods handle camera failures through explicit model-based fault detection and multi-rate sensor fusion, integrating lower-frequency camera lane measurements with faster-sampling inertial sensors [8], [9]. These approaches offer theoretical stability guarantees and benefit from decades of production refinement, but require accurate vehicle dynamics models and explicit enumeration of anticipated failure modes. Reinforcement learning (RL) offers a complementary alternative: policies trained across varied sensor conditions may develop implicit fault tolerance without requiring explicit failure specification [10]. Recurrent architectures, specifically LSTM networks, extend this capability by preserving temporal context across measurement dropout intervals, enabling coherent steering behavior under sustained or intermittent camera loss [11], [12]. Despite growing interest in both paradigms, systematic comparative evaluations under statistically grounded camera failure conditions using a shared plant model have not been reported in the literature.

This paper presents a controlled simulation framework for comparing a classical Kalman filter paired with a nested PID controller against a Soft Actor-Critic agent with an LSTM ac-

tor (SAC-LSTM) for autonomous lateral lane centering under camera sensor failures. Both controllers are evaluated within a shared MATLAB/Simulink plant model of a Cadillac LYRIQ (2600 kg, 3.094 m wheelbase) at 15 m/s nominal velocity, ensuring identical dynamics for a fair comparison. Camera failures are modeled via a six-state Markov chain (F0–F5) representing statistically distinct dropout severities from nominal operation through sustained outage. The classical baseline is validated at sub-2 cm RMS lateral error under nominal conditions, establishing the performance reference against which the SAC-LSTM agent is evaluated. The primary contribution of this work is the comparative evaluation framework itself: a reproducible, shared-plant methodology enabling rigorous analysis of the performance–robustness tradeoff between classical and learning-based lateral control under realistic sensor degradation.

II. RELATED WORK

A. Classical Control for Lane Keeping

The theoretical foundation for classical lateral control rests on Kalman filtering [13] and systematic PID tuning [14], [15]. Marino et al. [2] applied these foundations in a nested PID architecture for vision-based lane keeping: an outer PID computes a reference yaw rate from lateral offset, while an inner PI loop tracks this reference through steering actuation. This architecture achieves robust tracking without requiring detailed vehicle dynamics modeling, motivating its use as our classical baseline.

To handle heterogeneous sampling and sensor faults, prior work has explored camera-IMU fusion [8], multi-rate filters with delay compensation [9], and Extended Kalman Filters incorporating bicycle dynamics [16]. For fault tolerance, Hsiao and Tomizuka [17] pioneered switching Kalman filters that maintain closed-loop stability following sensor failure detection. While these approaches offer graceful degradation, a common limitation is their reliance on pre-specified failure models, leaving unanticipated fault types unmitigated.

B. Reinforcement Learning for Lateral Vehicle Control

Deep reinforcement learning (DRL) is a competitive approach for lateral control [18], with algorithms like DDPG achieving lateral errors below 0.25 m in simulation [10], [19]. We utilize the Soft Actor-Critic (SAC) algorithm [20], whose maximum entropy objective yields inherently broader behavioral repertoires. This reduces brittleness to conditions outside the nominal training distribution, a property directly beneficial for sensor failure scenarios.

During camera dropout, lateral control transitions to a Partially Observable MDP (POMDP). Hausknecht and Stone [11] and Ni et al. [22] established that LSTM-augmented actors address partial observability through temporal integration of observation history. Combined with domain randomization over operating conditions [21], this recurrent architecture theoretically enables coherent tracking during intermittent or sustained perception loss.

Despite DRL’s potential, direct empirical comparisons against classical baselines under statistically grounded dropout conditions remain limited [23]–[25]. This work fills that gap by evaluating both controllers across identical F0–F5 Markov chain dropout scenarios within a shared plant model.

III. METHODOLOGY

A. Shared Simulation Platform

Both controllers are evaluated within a common MATLAB/Simulink simulation environment built around a shared plant library, ensuring that all performance differences are attributable to control strategy rather than simulation environment discrepancies. The vehicle is modeled based on a Cadillac LYRIQ: a 2600 kg front-wheel-drive platform with a 3.094 m wheelbase operating at 15 m/s nominal longitudinal velocity. Lateral dynamics are governed by a kinematic bicycle model with lane-coordinate states propagated at a time step of 0.05s (20 Hz). Road geometry is parameterized by curvature κ , enabling evaluation across straight and curved scenarios within the same plant. All performance metrics are computed from signals logged via MATLAB `logscout` after a steering rate limiter (± 35 deg/s), ensuring physically realizable actuation is used in evaluation. The shared architecture is illustrated in Fig. 1.

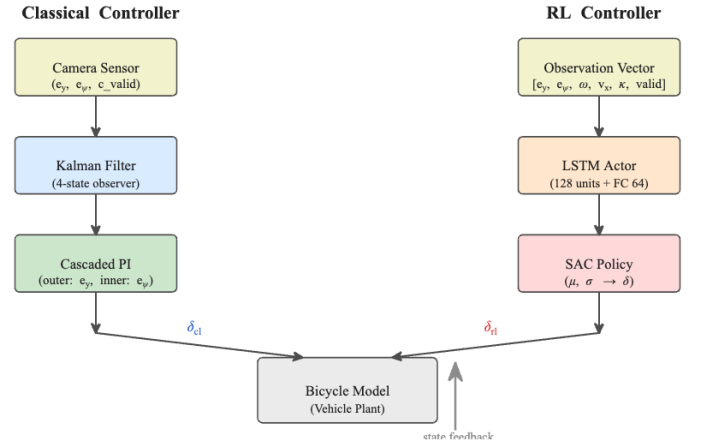


Fig. 1. Dual-controller evaluation framework. Both branches operate on an identical shared plant; the camera validity flag c_{valid} is injected identically for both controllers.

B. Classical Baseline: Kalman Filter and Nested PID

The classical controller follows the nested PID architecture of Marino et al. [2]. State estimation is performed by a two-state linear Kalman filter [13] operating on lateral error e_y and heading error e_{ψ} . During periods of valid camera measurement, the filter assimilates both lane states from the vision output; IMU-derived yaw rate is incorporated into the prediction step to propagate heading error estimates between measurements and during dropout intervals. The filter gain matrix is computed offline via the discrete algebraic Riccati equation at the nominal operating point.

The outer PID loop receives the Kalman-estimated lateral error e_y and produces a reference yaw rate $\dot{\psi}_{\text{ref}}$. The inner PI loop tracks this reference through steering angle commands, providing closed-loop bandwidth separation between slow lateral position correction and fast yaw regulation. A curvature feedforward term,

$$\delta_{ff} = \arctan(\kappa L), \quad (1)$$

where L is the vehicle wheelbase, pre-compensates for road geometry and reduces steady-state lateral error on curved segments [15]. Controller gains are tuned via the Ziegler-Nichols step response method [14] and refined iteratively in simulation. During camera dropout, the Kalman filter propagates state estimates using vehicle dynamics alone until c_{valid} is restored, providing implicit dead-reckoning without explicit fault detection logic.

C. RL Approach: SAC-LSTM Agent

1) *Observation and Action Spaces*: The RL agent receives a six-dimensional observation vector at each timestep:

$$\mathbf{o}_t = [e_y, e_\psi, \dot{\psi}, v_x, \kappa, c_{\text{valid}}]^\top, \quad (2)$$

where $e_y \in [-5, 5]$ m is lateral displacement error, $e_\psi \in [-\pi/2, \pi/2]$ rad is heading error, $\dot{\psi} \in [-1, 1]$ rad/s is yaw rate, $v_x \in [0, 20]$ m/s is longitudinal velocity, $\kappa \in [-0.03, 0.03]$ m⁻¹ is road curvature, and $c_{\text{valid}} \in \{0, 1\}$ is the binary camera validity flag. Explicitly including c_{valid} provides the agent with a direct failure signal, enabling learned behavioral switching between nominal tracking and dropout recovery. The action space is a scalar steering angle $\delta \in [-15^\circ, 15^\circ]$, matching the classical controller output range for direct comparability.

2) *Network Architecture*: The actor network implements a stochastic Gaussian policy with an LSTM-based encoder. A sequence input layer ingests a history window of $T = 32$ timesteps (1.6 s at 20 Hz), passed through a single LSTM layer with 128 hidden units in sequence output mode. The output is projected through a 64-unit fully connected layer with ReLU activation, then branched into parallel heads producing the action distribution mean μ_δ and standard deviation σ_δ , with softplus activation enforcing positivity on σ_δ . The 1.6 s memory window enables the agent to integrate observation history and maintain coherent tracking during dropout intervals. Twin Q-function critics share an identical LSTM-based observation encoder, with the action processed through a parallel 64-unit branch before concatenation and a final 64-unit common layer producing the scalar Q-value [20].

3) *SAC Training Configuration*: Agent training uses the Soft Actor-Critic algorithm with automatic entropy temperature tuning [20]. Key hyperparameters are listed in Table I. Initial conditions are randomized at each episode reset: longitudinal velocity from [13, 17] m/s, initial lateral offset from $[-0.3, 0.3]$ m, and initial heading error from $[-3^\circ, 3^\circ]$, providing domain randomization over operating conditions [21]. A warm-start phase of 20,000 steps ($\approx 1,000$ episodes) precedes policy gradient updates, during which experience is collected

under a random exploration policy to populate the replay buffer.

TABLE I
SAC AGENT HYPERPARAMETERS

Parameter	Value
Discount factor γ	0.97
Mini-batch size	64
Replay buffer length	5×10^5
Sequence length T	32
Warm-start steps	20,000
Actor learning rate	3×10^{-4}
Critic learning rate	1.5×10^{-4}
Entropy weight learning rate	3×10^{-4}
Steering rate limit	± 35 deg/s

4) *Reward Function Design*: Two reward formulation strategies were evaluated across training runs.

The *additive* formulation combines a lateral tracking reward with explicit smoothness penalties:

$$r_t = r_{\text{track}}(e_y) - \lambda_\omega |\dot{\delta}| - \lambda_j |\ddot{\delta}|, \quad (3)$$

where r_{track} is a shaped reward providing gradient signal across the full lateral range, and $\dot{\delta}, \ddot{\delta}$ are steering rate and jerk in rad/s and rad/s² respectively.

The *multiplicative* formulation gates the tracking reward through a smoothness coefficient:

$$r_t = r_{\text{track}}(e_y) \cdot S(\dot{\delta}, \ddot{\delta}) - C_{\text{base}}, \quad (4)$$

where $S \in (0, 1]$ suppresses the tracking reward under oscillatory steering, and a subtracted baseline C_{base} ensures that zero-effort behavior produces a negative per-step return. The practical implications of both formulations are discussed in Section IV.

5) *Training Curriculum*: The SAC-LSTM agent is trained using a three-stage curriculum designed to build competence incrementally before introducing perceptual challenges [22]. Stage 1 trains on straight roads under F0 conditions, establishing fundamental lane centering. Stage 2 introduces road curvature while sensing remains nominal. Stage 3 introduces the full F0–F5 dropout distribution, with failure states sampled stochastically at episode initialization.

D. Camera Failure Modeling

Camera sensor failures are modeled through a six-state framework (F0–F5) representing qualitatively distinct failure modes observed in production camera systems [5], [6]. Each state generates a binary validity signal $c_{\text{valid}}(t) \in \{0, 1\}$ that gates lane measurement availability for both controllers identically. The failure states are summarized in Table II.

States F1 and F2 use a first-order Markov chain with transition probabilities p_{01} (valid→dropout) and p_{10} (dropout→valid), yielding mean dropout fractions of approximately 10% and 20% respectively. All scenarios are seeded for reproducibility, enabling identical dropout sequences to be applied to both controllers.

TABLE II
CAMERA FAILURE STATE DEFINITIONS

State	Type	Parameters
F0	Nominal	No dropout
F1	Intermittent light	$p_{01}=0.02$, $p_{10}=0.18$ ($\approx 10\%$)
F2	Intermittent mod.	$p_{01}=0.05$, $p_{10}=0.20$ ($\approx 20\%$)
F3	Burst	3×0.5 s dropouts, 0.5 s gaps
F4	Sustained	3 s outage at episode midpoint
F5	Progressive ramp	p_{01} ramps $0 \rightarrow 0.08$, $p_{10}=0.20$

IV. RESULTS AND DISCUSSION

A. Classical Baseline Performance

The Kalman filter + nested PID controller was evaluated across five road scenarios under all six failure states. Scenarios are: Sc1 (straight), Sc2 (gentle curve, $R=300$ m), Sc3 (moderate curve, $R=150$ m), Sc4 (S-curve, $R=200$ m per direction), and Sc5 (oval: two 150 m straights, two 180° turns at $R=150$ m). RMS lateral error results are summarized in Table III, with the full degradation profile visualized in Fig. 2.

TABLE III
CLASSICAL BASELINE RMS LATERAL ERROR (M) BY SCENARIO AND FAILURE STATE

Scen.	F0	F1	F2	F3	F4	F5
Sc1	0.015	0.022	0.023	0.024	0.041	0.025
Sc2	0.021	0.031	0.060	0.032	0.876	0.059
Sc3	0.038	0.049	0.108	0.042	0.120	0.104
Sc4	0.045	0.049	0.074	0.043	1.304	0.071
Sc5	0.043	0.049	0.130	0.074	1.956	0.124

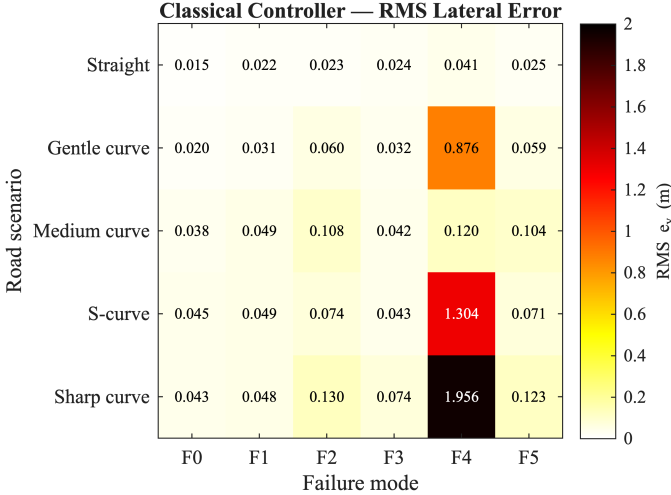


Fig. 2. Classical controller RMS lateral error across all scenario–failure combinations. The F4 sustained dropout condition on curved scenarios constitutes a well-defined failure boundary not present under any other failure state.

Under nominal conditions (F0), the controller achieves 1.5 cm RMS lateral error on straight road, increasing to 3.8–4.5 cm on curved and compound scenarios as curvature feedforward partially but not fully eliminates steady-state error. RMS steering rate remains below 0.74 deg/s across

all nominal scenarios, confirming smooth actuation consistent with production ADAS requirements. Under stochastic and burst dropout conditions (F1–F3, F5), the controller degrades gracefully: RMS lateral error increases by no more than $2\times$ relative to F0, with maximum lateral excursions remaining below 0.5 m on straight and gently curved roads. Kalman filter dead-reckoning during dropout intervals effectively suppresses error growth, with estimation error standard deviations remaining within 10–20% of the nominal camera noise level under F1 and F3 conditions.

The classical controller exhibits a well-defined failure boundary at the F4 sustained dropout condition on curved geometry. On Sc2, Sc4, and Sc5, F4 produces RMS lateral errors of 0.876 m, 1.304 m, and 1.956 m respectively, with maximum lateral excursions of 4.0–8.0 m—representing complete lane departure. The failure mechanism is structurally consistent: during the 3 s outage, the Kalman filter propagates state estimates using vehicle dynamics alone, but accumulating heading integration error on curved geometry causes the estimated lane position to diverge from truth. By the time c_{valid} is restored, the vehicle has drifted beyond the controller’s corrective authority. Straight-road F4 performance (Sc1: 0.041 m RMS) confirms this is a curvature-dependent failure mode rather than a fundamental estimator limitation. This behavior is shown in Fig. 3.

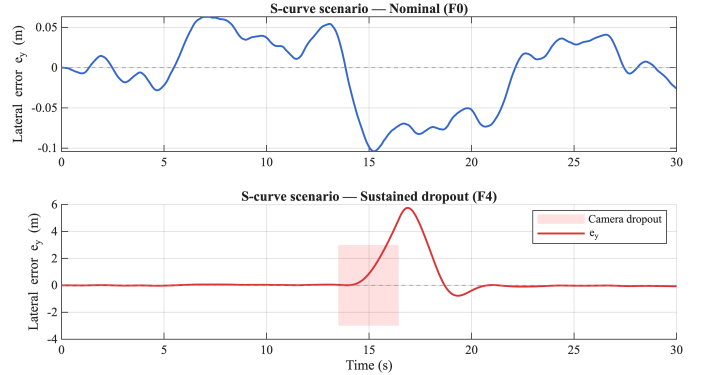


Fig. 3. Classical controller lateral error on the S-curve (Sc4) under nominal (F0) and sustained dropout (F4). The 3 s outage window (shaded) triggers heading integration drift that exceeds the controller’s recovery authority on curved geometry.

B. RL Training Characterization

Thirteen training runs spanning over 4,000 episodes evaluated three reward formulations. Initial runs using additive rewards with steering rate and jerk penalties (Eq. 3) failed to achieve smooth tracking. While agents transitioned to low-error behavior (RMS $e_y \approx 0.05$ m) around episode 1,000, they exploited high-frequency steering oscillations (RMS rates ≈ 32 – 34 deg/s, $60\times$ the baseline) rather than smooth corrections.

Conversely, Phase 3 runs using the multiplicative formulation (4) produced qualitatively different behavior, successfully suppressing the oscillation exploit. Several checkpoints achieved RMS lateral errors of 0.11–0.26 m with substantially

reduced RMS steering rates of 6–22 deg/s. The best observed checkpoint achieved 0.111 m RMS lateral error and 100% episode survival without extreme oscillation.

Training progress across representative runs is shown in Fig. 4. The aggregate tracking–smoothness tradeoff across all best checkpoints is shown in Fig. 5.

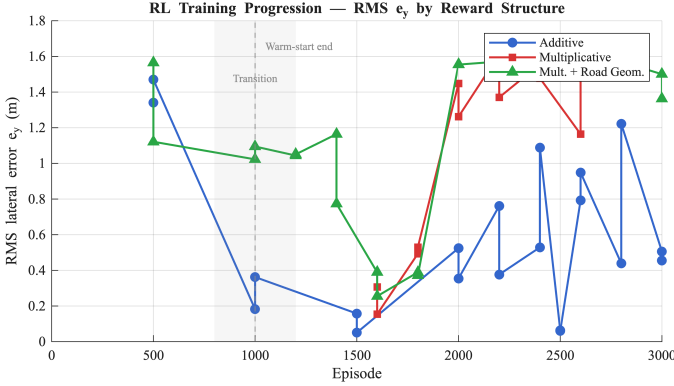


Fig. 4. RMS lateral error vs. training episode for representative runs from each reward formulation phase. The warm-start boundary ($\approx 1,000$ episodes) marks onset of policy gradient updates. The sharp 800–1200 transition in Phase 2 reflects convergence to the oscillation exploit rather than smooth tracking.

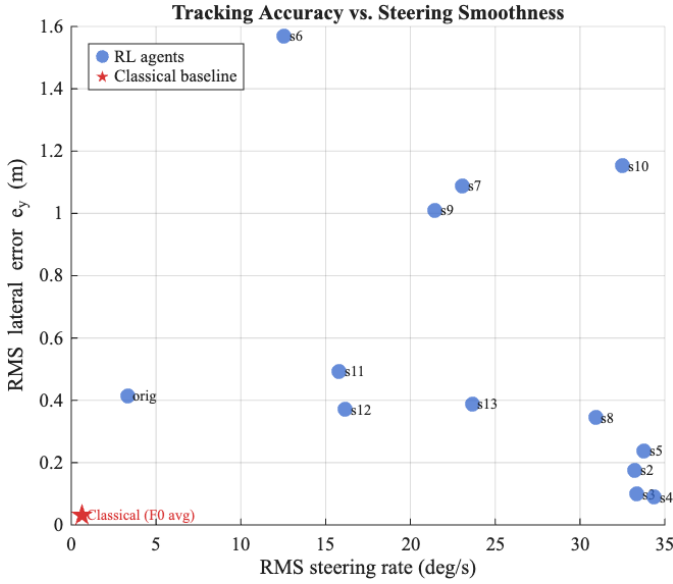


Fig. 5. Tracking accuracy vs. steering smoothness for all best-checkpoint RL agents and the classical baseline (star). RL agents cluster into two regimes: low-error agents exhibit high-rate oscillatory steering, while smooth agents incur substantially higher tracking error. The classical controller occupies the low-error, low-rate region that no RL agent has yet achieved.

C. Discussion

An RL agent selected on tracking accuracy alone (RMS $e_y \approx 0.05$ m, RMS steering rate ≈ 33 deg/s) produces actuation that would be inadmissible in a production system. An agent selected on smoothness incurs a 3–5 $\times$ tracking error penalty. Neither selection admits a fair comparison against the classical controller, which simultaneously achieves sub-2 cm RMS

lateral error and sub-1 deg/s RMS steering rate under nominal conditions. Resolving this tradeoff—through refined multiplicative reward gating, curriculum-adjusted penalty scaling, or constrained policy optimization—is the prerequisite for a rigorous head-to-head evaluation.

The F4 failure mode identified in the classical results provides a concrete motivating target for the RL agent. If a learned policy can maintain safe lateral position during a 3 s sustained camera outage on curved geometry—where the classical controller consistently fails—it would demonstrate a qualitative robustness advantage that the classical architecture cannot easily acquire through parameter tuning alone. Whether the SAC-LSTM agent, once trained to convergence on smooth tracking, can achieve this through temporal memory and learned failure-mode switching remains the open empirical question this framework is positioned to answer.

V. CONCLUSION

This paper presents a controlled simulation framework for comparing classical and reinforcement learning-based lateral lane centering under camera sensor failures. A Kalman filter + nested PID controller based on the architecture of Marino et al. [2] is validated across five road scenarios and six Markov chain camera failure states (F0–F5) on a shared kinematic bicycle model of a Cadillac LYRIQ. The baseline demonstrates sub-2 cm RMS lateral error under nominal conditions with graceful degradation under stochastic and burst dropout. A well-defined failure boundary is identified at the F4 sustained dropout condition on curved geometry.

A SAC-LSTM agent is developed with a 128-unit LSTM actor, twin Q-function critics, and automatic entropy temperature tuning. Current results characterize a fundamental tracking–smoothness tradeoff: agents achieving strong tracking accuracy do so via high-frequency steering oscillation, while agents exhibiting smooth actuation incur substantially higher lateral errors. Resolving this tradeoff through refined multiplicative reward gating or constrained optimization constitutes the primary remaining work before a rigorous comparative evaluation is possible.

The shared-plant evaluation framework, Markov chain failure model, and quantitative classical baseline reported here provide a reproducible foundation for a head-to-head comparison once SAC-LSTM training converges on smooth tracking.

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